Strong genotype-by-genotype and genotype-by-environment interactions induced by antibiotic resistance mutations

Alicia H. Williams, Ian Gooi, Madelaine Boggan, Ziqing Liu, Andrew D. Letten, and Jan Engelstädter

School of the Environment, The University of Queensland, Brisbane, Queensland 4072, Australia

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Abstract

The parallel evolution of antibiotic resistance mutations offers a promising basis for predicting evolutionary change. However, the extent to which parallel genetic evolution is reflected at the phenotypic level remains unclear. We investigated how genetic background and temperature influence the fitness effects of homologous rifampicin resistant mutations in absence of antibiotics across two temperatures and three species (*Escherichia coli*, *Acinetobacter baylyi* and *Lactiplantibacillus plantarum*). We measured the effect of resistance mutations on two growth traits (maximum growth rate and maximum optical density) across all conditions and fit Bayesian linear models to this data. Our model fits provide evidence for pervasive genetic interactions in both traits, including mutation by species (GxG), genotype by temperature (GxE), and GxGxE interactions. In particular, the effect of resistance mutations was strongly dependent on genomic background, to the extend that mutations often reduced growth in one species but increased it in another (sign epistasis). Similarly, the effect of mutations on growth was strongly influenced temperature, with some mutations increasing growth at one but decreasing growth at the other temperature. Most notably, three resistance mutations induced strong heat sensitivity in *A. baylyi*. Our results support previous studies that report strong epistatic and GxE effects of resistance mutations, indicating that in the absence of additional data and more mechanistic models, predicting fitness effects of resistance mutations is very challenging.

Introduction

Owing to theoretical advances and technological innovation, recent years have seen a surge in interest in predicting evolutionary change (Lassig et al. 2017; Wortel et al. 2023). Accurate predictions could benefit fields in which evolution is rapid and its impacts significant, including medicine, public health, agriculture, and biological conservation. In particular, forecasting antibiotic resistance evolution in bacteria has received considerable attention (Schenk and de Visser 2013; Pinheiro 2024; Rolff et al. 2024). This is not only because of the stakes involved given the emerging resistance crisis, but also because resistance mutations hold promise for successful predictions, as antibiotic resistance often has a relatively simple genetic basis that is highly conserved across species (Wong and Kassen 2011). This is especially true for mutations at the antibiotic target site that confer resistance by preventing antibiotic binding (Blair et al. 2015). Since antibiotic targets are usually proteins involved in core cellular functions common to all bacteria (e.g., protein synthesis), resistance often evolves via homologous mutations in different species. This parallelism offers the promise of predictions that are robust with regard to species or strain.

Resistance to rifampicin (RIF) is one such case. RIF is an antibiotic that has been used since the 1960s to treat a wide range of infections and is a first-line drug to treat tuberculosis (Goldstein 2014; Adams et al. 2021). RIF binds to the β-subunit of the RNA polymerase (RNAP), blocking elongation of newly formed transcripts. Resistance to rifampicin typically evolves through point mutations in the gene *rpoB* that codes for the RNAP β-subunit. Early mutant screens in *E. coli* demonstrated that these mutations typically arise at specific positions in *rpoB* that lead to amino acid changes close to the binding site of RIF (Jin and Gross 1988), and subsequent work has demonstrated that the same (i.e., homologous) mutations also confer RIF resistance in *Mycobacterium tuberculosis* and many other species (e.g. Morlock et al. 2000; Goldstein 2014). Given the broad phylogenetic distribution of these species, RIF resistance evolution presents a remarkable case of parallel evolution across the entire bacterial tree of life (Bolourchi et al. 2025).

It is less clear to what extent parallel evolution of antibiotic resistance at the genetic level is reflected at the phenotypic level. Antibiotic resistance is typically associated with fitness effects in the absence of drugs, measured for example as a change in the growth rate or competitive fitness relative to the ancestral genotype. (This is often referred to as the 'cost of resistance', but here we follow Lenormand et al. (2018) and use the more neutral term 'fitness effects in absence of drugs'.) Our ability to predict the evolution of antibiotic resistance depends on these fitness effects being similar across different genetic backgrounds (different strains or different species), and also across different environments. However, there is mounting evidence that fitness effects can vary significantly by both genetic background (Vogwill et al. 2016; Wong 2017; Vogwill and MacLean 2015; Vogwill et al. 2014; Trindade et al. 2009; Apjok et al. 2019; Hinz et al. 2024; Rodríguez-Verdugo et al. 2013; Borrell et al. 2013) and environment (Clarke et al. 2020; Leehan and Nicholson 2021, 2022; Gifford et al. 2016; Mira et al. 2022; Ghenu et al. 2023). Variation of fitness effects by genetic background can be considered a case of epistasis, also referred to as genotype-by-genotype (G×G) interactions. Epistasis is defined as a deviation from independent genetic effects (here, effects of resistance mutations and species/strain on fitness). Independence can be defined on either an additive or multiplicative scale. In extreme cases, deviations from independence can be so pronounced that not only the magnitude of a phenotypic effect depends on the genetic background (magnitude epistasis) but also its direction (sign epistasis; Weinreich et al. 2005). Variation of fitness effects in different environments are referred to as a genotype-by-environment (G×E) interactions, considered by some authors as a special case of pleiotropy (but see Lenormand et al. 2018). Conceptually, GxE effects are analogous to GxG effects and we can again distinguish between magnitude and sign GxE effects (or pleiotropy). A final complication occurs when epistasis itself is dependent on the environment (or, equivalently, GxE effects depend on genetic background), giving rise to GxGxE interactions.

Here we investigated the impact of genetic background and temperature on the fitness effects of RIF resistant mutants in the absence of antibiotics. We measured growth effects of rifampicin resistance mutations across three species, *Escherichia coli*, *Acinetobacter baylyi* and *Lactiplantibacillus plantarum*, and at two temperatures, 30 °C and 37 °C. We then fit a series of Bayesian linear models to our data, demonstrating strong GxG (epistasis), GxE, and GxGxE effects in this system.

Methods

Strains and culture methods

The species used were *Escherichia coli* MG1655, *Acinetobacter baylyi* ADP1 and *Lactiplantibacillus plantarum*. Aside from these ancestral strains (henceforth referred to as wildtype), we used a total of 55 rifampicin resistant mutants across these three species that were obtained in prior mutant screens (Gooi et al., in preparation). *E. coli* and *A. baylyi* were grown in Luria-Bertani (LB) broth, and *L. plantarum* (which is unable to grow in LB) was grown in De Man–Rogosa–Sharpe (MRS) broth. Unless specified, species were incubated in their recorded optimal temperature; 37°C for *E. coli* and 30°C for *A. baylyi* and *L. plantarum*. Stocks of all strains and mutants were stored at -80°C in 15% glycerol within the preferred media for the species. Overnight cultures used for the following experiments were grown from either a scraping of the -80°C glycerol stock or picked from a colony on LB agar spread from the -80°C glycerol stock.

Growth assays

We performed growth assays of the three wildtypes and all mutants at each species' preferred temperature. In addition, for the *E. coli* and *A. baylyi* wildtype and all mutants with homologous RIF resistance mutations across those two species we also measured growth at the other species' preferred temperature (30 °C and 37 °C, respectively). For each growth assay, a 96 well plate with 178µL of growth media was inoculated with 2µL of overnight cultures. Plates were incubated under continuous shaking for 24 hours in a microplate reader (BioTec Synergy H1 or Epoch) and the absorbance optical density at 600 nm (OD_{600}) was measured every 5 minutes. Growth assays were performed with six replicates for each wildtype and mutant in their preferred temperature, and with three replicates for the additional growth assays in the non-preferred temperature.

Statistical analysis

Growth curves of all conditions from the absorbance OD_{600} measures were analysed using the in-house R package *grow96* (https://github.com/JanEngelstaedter/grow96). Two growth traits were estimated for each growth curve: maximum OD (maxOD) and the maximum per capita growth rate ($\mu_{\max}$). We used the easylinear method (Hall et al. 2014) as implemented in the *growthrates* package (Petzold 2022) to estimate maximum growth rate.

We fit linear models to the data, with either $\mu_{\max}$ or maxOD as the response variable, and species, temperature and mutation as the independent variables. These models included: 1) models with only species and mutation as independent variables (at each species' preferred temperature); 2) models with only temperature and mutation as independent variables (separately for each species); and 3) full models with all three independent variables for the *E. coli* / *A. baylyi* species pair. We used STAN in conjunction with the R package *brms* (Bürkner 2017) for Bayesian model fitting, assuming improper flat priors for all model coefficients. Posteriors were then extracted from the fitted models, and the mean and 95% credible intervals (as highest density intervals) were determined from the posteriors.

Results

Overview

We conducted growth assays of populations grown in monoculture and absence of drugs for a total of 55 RIF resistant mutants across three species (Table 1), as well as the three susceptible wildtype genotypes. From each growth curve, we estimated the maximum per captita growth rate ($\mu_{\max}$) and the max OD. A substantial fraction of the mutations in our panel were homologous between species, allowing us to estimate interactions between mutations and genetic background (GxG interactions, or "species-level epistasis"). For the mutations that overlapped between *E. coli* and *A. baylyi* we measured growth at two temperatures (30 °C and 37 °C) so as to also estimate interactions between mutations and temperature (GxE interactions).

Figure S1 shows the $\mu_{\max}$ and max OD estimates across the three species and two temperatures. In all three species, most resistance mutations did not have a strong effect on growth rate, although some mutations reduced growth considerably (e.g. mutations at position 529 in *E. coli*). As expected, *E. coli* grew much slower and to lower densities at 30 °C than at 37 °C. However, in *A. baylyi*—normally grown at 30 °C—increasing temperature to 37 °C had a negative effect only on max OD but not on $\mu_{\max}$.

We used Bayesian inference to fit three types of linear models to our data, corresponding to the three types of genetic interactions: 1) models of the form Trait $\sim$ mutation $*$ species, where Trait can be either $\mu_{\max}$ or maxOD, to estimate GxG effects, 2) models of the form Trait $\sim$ mutation $*$ temperature to estimate GxE effects, and 3) models of the form Trait $\sim$ mutation $*$ species $*$ temperature to estimate GxG, GxE and GxGxE effects. We visualise the fits for the former two models in plots that make it easy to see both the magnitudes and types of GxG or GxE interactions (see Figure 1 for a schematic and explanation of these plots). These types include 1) magnitude epistasis (or magnitude GxE interactions), where there are positive or negative deviation from additivity but where the direction of both effects relative to a reference genotype is the same irrespective of the state at the other locus or environment, 2) sign epistasis, where the direction of one but not the other effect differs depending on the state of the other locus or environment, and 3) reciprocal sign epistasis, where the direction of both effects differs depending on the other locus or environment.

Position	Mutation	E. coli	A. baylyi	L. plantarum
146	V $\rightarrow$ G	146	148	
146	V $\rightarrow$ F	146		
509	S $\rightarrow$ R	509	516	
511	L $\rightarrow$ S		518	
511	L $\rightarrow$ P	511		
512	S $\rightarrow$ P	512	519	
512	S $\rightarrow$ F	512		
513	Q $\rightarrow$ P	513		477
513	Q $\rightarrow$ K			477
516	D $\rightarrow$ A		523	
516	D $\rightarrow$ G	516	523	
516	D $\rightarrow$ N	516		480
516	D $\rightarrow$ E			480
516	D $\rightarrow$ Y			480
518	N $\rightarrow$ D		525	
522	S $\rightarrow$ F	522		
526	H $\rightarrow$ D	526	533	
526	H $\rightarrow$ Y	526	533	490
526	H $\rightarrow$ N	526	533	490
529	R $\rightarrow$ G	529		493
529	R $\rightarrow$ C	529		493
529	R $\rightarrow$ L	529		
529	R $\rightarrow$ H			493
531	dup		538	
531	S $\rightarrow$ L		538	495
531	S $\rightarrow$ F	531		
533	L $\rightarrow$ S		540	
533	L $\rightarrow$ W		540	
534	G $\rightarrow$ E			498
537	G $\rightarrow$ D		544	
564	P $\rightarrow$ L		571	528
572	I $\rightarrow$ L	572	579	
572	I $\rightarrow$ S	572	579	
572	I $\rightarrow$ F	572		
574	S $\rightarrow$ L		581	

Table 1: Rifampicin resistance mutants used in this study. The first two columns show the amino acid position in *E. coli* coordinates and the amino acid substitution, respectively. The remaining columns show the mutations used in this study, with numbers indicating the amino acid positions in the three focal species. Mutations shown in the same row are homologues.

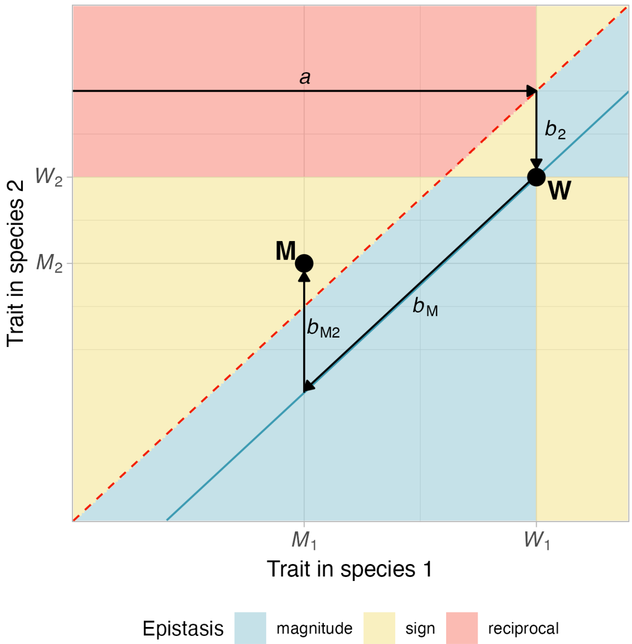

Figure 1: Schematic of different types of epistasis between species and resistance mutations and their relationship to linear model fits. The two axes in this plot show the values of a trait (e.g. growth rate) in species 1 and 2, respectively. The wildtype strain, carrying no resistance mutation, labelled "W", is characterised by trait values W_1 and W_2 in the two species. The different areas in this plot indicate different types of epistasis (magnitude, sign and reciprocal sign epistasis) for mutants situated in these areas. The blue line indicates additive trait effects (no epistasis), and the areas above and below the blue line are characterised by positive and negative epistasis, respectively. The dashed red line indicates equal trait values of a genotype in both species. An example mutant genotype, labelled "M", is shown that has trait values M_1 and M_2 in the two species. This mutant is characterised by positive (non-reciprocal) sign epistasis. This is because the mutation reduces the trait value in both species ($M_1 < W_1$ and $M_2 < W_2$) but the direction of the trait change between species 1 and 2 depends on the mutation ($W_2 < W_1$ but $M_2 > M_1$). The black arrows correspond to coefficients in a linear model fitted to data. The intercept a represents the baseline trait of the species 1 wildtype. Coefficient b_2 captures the effect of switching from species 1 to species 2, so that $a+b_2$ is the trait value for the species 2 wildtype. Coefficient b_M is the effect of the mutation, so that $a + b_\mathrm{M}$ is the trait value of the species 1 mutant. Finally, b_{M2} is the coefficient capturing the interaction between species 2 and the mutation, so that $a + b_2 + b_M + b_{M2}$ gives the trait value of the species 2 mutant. Note that "species" can be replaced by "environment" in this figure, which would then show analogous GxE instead of GxG interactions.

Epistasis across species

Our GxG model fit reveals that epistatic interactions between RIF resistance mutations and genetic background (species) are abundant but not ubiquitous (Figure 2). For both species comparisons involving *E. coli* we see a wide spectrum of epistatic interactions affecting maximum growth rate, ranging from negative to near zero (additive effects) to positive. For example, in the *E. coli* vs. *L. plantarum* comparison (Figure 2B), mutations R529G and R529C are characterised by positive epistasis (faster growth in *L. plantarum* relative to the expectation under additive effects, given growth rates of the same mutation in *E. coli* and the two wildtypes), mutation H526Y has zero epistasis, and mutation Q513P has negative epistasis. Epistatic effects for maximum growth rate also vary considerably, although their sign appears more consistent in each species comparison. Both magnitude epistasis and, in some cases, strong sign epistasis was observed. However, sign epistasis was always unidirectional and never reciprocal. In some cases the effects of mutations at the same site (H526D vs. H526Y and R529C vs. R529G) are very similar, but this is not universally true. Overall, there does not seem to be a clear pattern for the distribution of epistatic interactions across mutations and growth traits.

Interactions between mutations and temperature

We next investigated genetic interactions between mutations and temperature (GxE interactions), for both growth traits and the two species *E. coli* and *A. baylyi*. Figure 3A shows that in *E. coli* there are no or very weak GxE interactions involved in $\mu_{\max}$, as seen by the overlap between the credible intervals and the blue diagonal indicating additive effects. In contrast, most mutations show strongly positive GxE effects in maxOD in *E. coli*, with many of the values falling within the domain of reciprocal sign GxE effects (red area in Figure 3C). This is because even though maxOD is considerably lower in wildtype *E. coli* at 30 °C than at 37 °C and is generally lower for mutants than for wildtype at 37 °C, many mutants grow to higher maxOD than both the wildtype at 30 °C and the same mutant at 37 °C.

In *A. baylyi*, there is a clear pattern of three mutations (S509R, S512P and H526D) showing strongly reduced growth at 37 °C compared to 30 °C, both in terms of $\mu_{\max}$ and maxOD (Figure 3B and D). These three mutations are also characterised by positive sign GxE effects because their growth is increased much more when switching from 37 °C to 30 °C than expected under additive effects, given the corresponding growth of the wildtype at the two temperatures. In

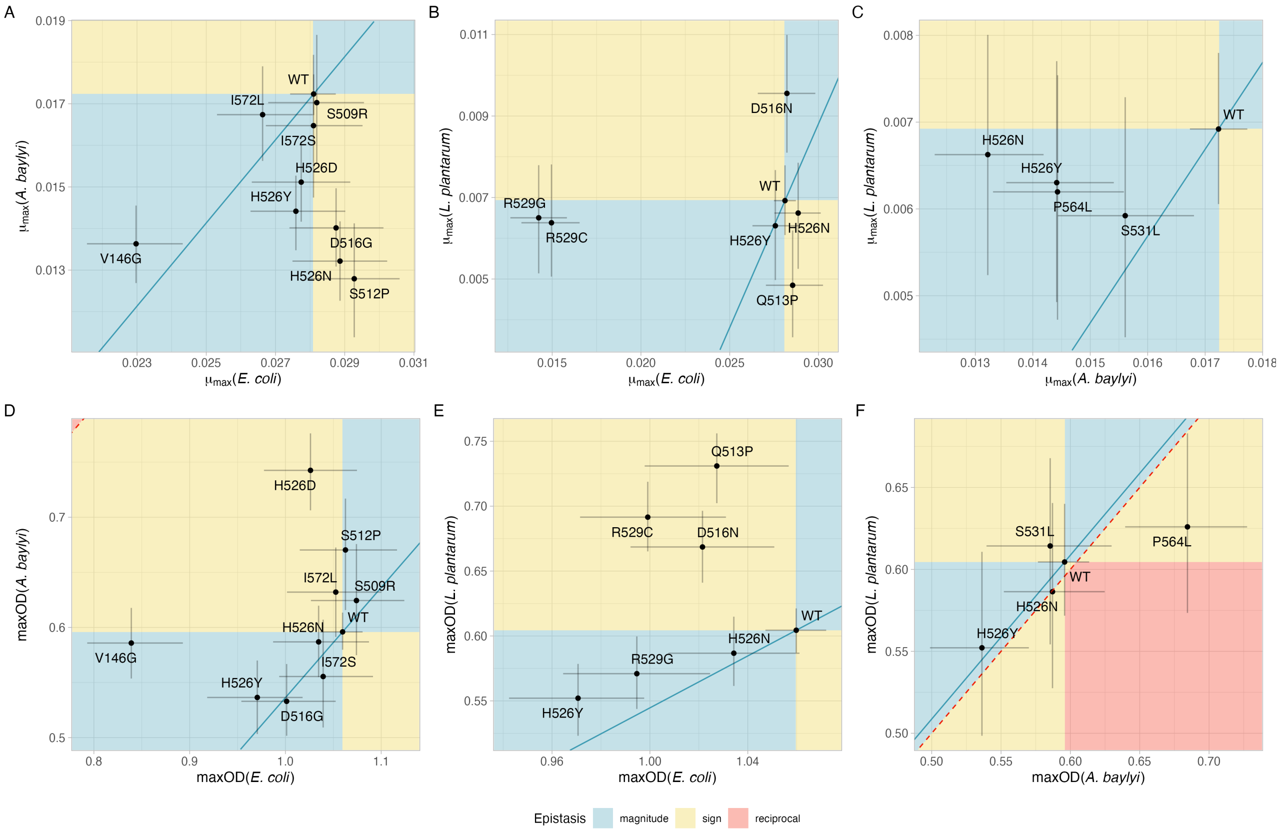

Figure 2: G×G interactions between rifampicin resistance mutations and species. The three columns show comparisons between three species pairs: *E. coli* vs. *A. baylyi* (A, D), *E. coli* vs. *L. plantarum* (B, E) and *A. baylyi* vs. *L. plantarum* (C, F). The two rows show the two growth traits measured: maximum growth rate ($\mu_{\max}$, A, B, C) and maximum OD (D, E, F). In each plot, the two axes represent the estimated growth parameter for the two species. Each point shows the estimated mean of the posterior distribution of a mutant, and error bars show 95% credible intervals. The background colour denotes areas with different types of G×G interactions (magnitude, sign, or reciprocal sign epistasis). The blue lines indicates the absence of any additive epistasis (additive effects only). The dashed red line is the main diagonal and indicates equal growth trait across the two species.

other words, those three mutations render the bacteria particularly heat sensitive. By contrast, the other resistance mutations in *A. baylyi* exibit no or very weak GxE effects in either growth trait.

G×G×E effects between mutations, species and temperature

We next fitted models for $\mu_{\max}$ or max OD as a function of mutation, genetic background (species), temperature and their three-way interaction, enabling us to look at the same data as in the previous section from the angle of a full linear model. Figure 4 shows the estimated effects of the various coefficients in these model fits. These coefficients reflect estimates of epistasis (GxG), GxE effects and GxGxE effects.

The first three coefficients show the effects of species and temperature in the wildtype. The coefficients for *A. baylyi* and a temperature of 30 °C are negative for both $\mu_{\max}$ and maxOD, reflecting the fact that *E. coli* at 37 °C grows better than *A. baylyi* at 37 °C and better than at 30 °C. However, the negative growth effects of *A. baylyi* at 37 °C are largely compensated by the positive interaction coefficient between species and temperature (AB:T30 in Figure 4).

The effects of individual mutations, by themselves, are generally weak, with the only notable exception being V146G. This is also apparent in Figure S1 showing that most mutations (that overlapped between *E. coli* and *A. baylyi*) had only weak growth effects. Mutation by temperature interactions are also weak for $\mu_{\max}$, mirroring the absence of strong GxE interactions in *E. coli* (compare Figure 3A). By contrast, there are strong mutation by species interactions for both $\mu_{\max}$ and maxOD.

Our full model also allows us to estimate GxGxE interactions, demonstrating that such interactions are present for several RIF resistance mutations. These effects reflect the strong temperature sensitivity induced by the three mutations highlighted in the previous section (S509R, S512P and H526D), with the positive GxGxE coefficients for these three mutations compensating for the strongly negative coefficients describing growth at 37 °C. To further verify the importance of GxGxE effects, we also fit a linear model that did not include the three way interaction but still included all two-way interactions (Trait $\sim$ mutation $*$ species + mutation $*$ temperature + species $*$ temperature instead of Trait $\sim$ mutation $*$ species $*$ temperature. We compared the marginal likelihoods of the full and reduced model (obtained through bridge sampling). This yielded a Bayes factor of 1.93×10^7, indicating "decisive support" for the full

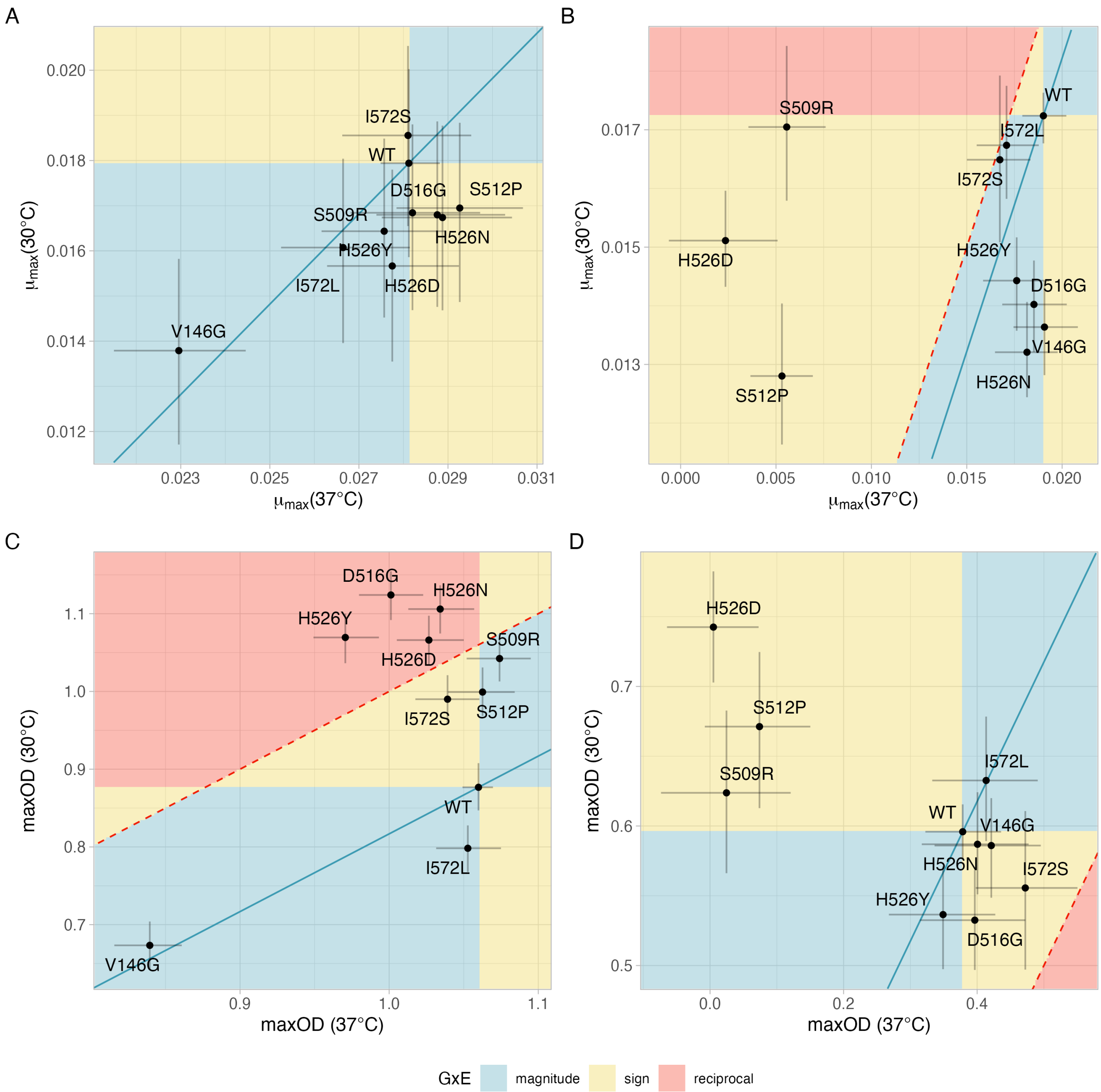

Figure 3: G×E interactions between rifampicin resistance mutations and temperature. The four panels correspond to the two species *E. coli* (A and C) and *A. baylyi* (B and D) as well the two growth traits maximum growth rate ($\mu_{\max}$, A, B) and maximum OD (C, D). In each plot, the represent the estimated growth parameter at 30 °C and 37 °C, respectively. Each point shows the estimated mean of the posterior distribution of a mutant, and error bars show 95% credible intervals. The background colour denotes areas with different types of G×E interactions (magnitude, sign, or reciprocal sign G×E interactions). The blue lines indicates the absence of any G×E interactions (additive effects only). The dashed red line is the main diagonal and indicates growth traits across the two temperatures. See Fig. 1 for more guidance on the interpretation of this figure.

model on Jeffreys' scale.

Discussion

We measured the impact of RIF resistance mutations on growth traits (maximum growth rate and maximum optical density in monoculture) in three species (*E. coli*, *A. baylyi* and *L. plantarum*) and two temperatures (30 °C and 37 °C). Our results indicate strong GxG effects between mutation and genetic background (including sign epistasis) as well as strong GxE and GxGxE effects that characterise this 'fitness seascape'.

RIF resistance mutations in *rpoB* have been identified and characterised phenotypically in many species, including *Mycobacterium* spp., *E. coli*, *Bacillus* spp. and many others (e.g. Jin and Gross 1988; Vogler et al. 2002; Gagneux et al. 2006; Trindade et al. 2009; Goldstein 2014; Leehan and Nicholson 2021; Yang et al. 2023; Barilar et al. 2024; Bolourchi et al. 2025). It is clear from this large body of work that RIF resistance mutations themselves are generally highly conserved. However, to our knowledge only one previous study has directly compared fitness effects of RIF resistance mutations in the absence of RIF across species (Vogwill et al. 2016). Here, the authors measured the competitive fitness effects of homologous RIF resistance mutations in seven *Pseudomonas* species, reporting that those effects varied considerably across species: almost 40% of the variance in fitness was explained by interactions between mutations and species, and very little by mutation and species as individual effects. Even within a single species, the fitness effects of *rpoB* mutations can be highly variable, as shown by Hinz et al. (2024) for identical resistance mutations (conferring resistance to rifampicin, ciprofloxacin and streptomycin) across different clinical isolates of *E. coli*. Similarly, Trauner et al. (2021) showed that fitness effects of the S531L mutation vary markedly in different strains of *M. tuberculosis*. Knopp and Andersson (2018) appear to contradict these results, reporting predictable growth phenotypes for 13 chromosomal resistance mutations to different antibiotics across ten strains of *Salmonella enterica* and *E. coli* MG1655. However, the single RIF resistance mutation in their panel (S531L) did in fact exhibit considerable variation in its relative fitness effects in this study, even between strains that were almost identical. Overall, our results of strong epistatic effects across species are therefore not surprising, given that the three species we studied are only very distantly related (*E. coli* and *A. baylyi* are both gammaproteobacteria but belong to different orders, whereas *L. plantarum* belongs to a different phylum).

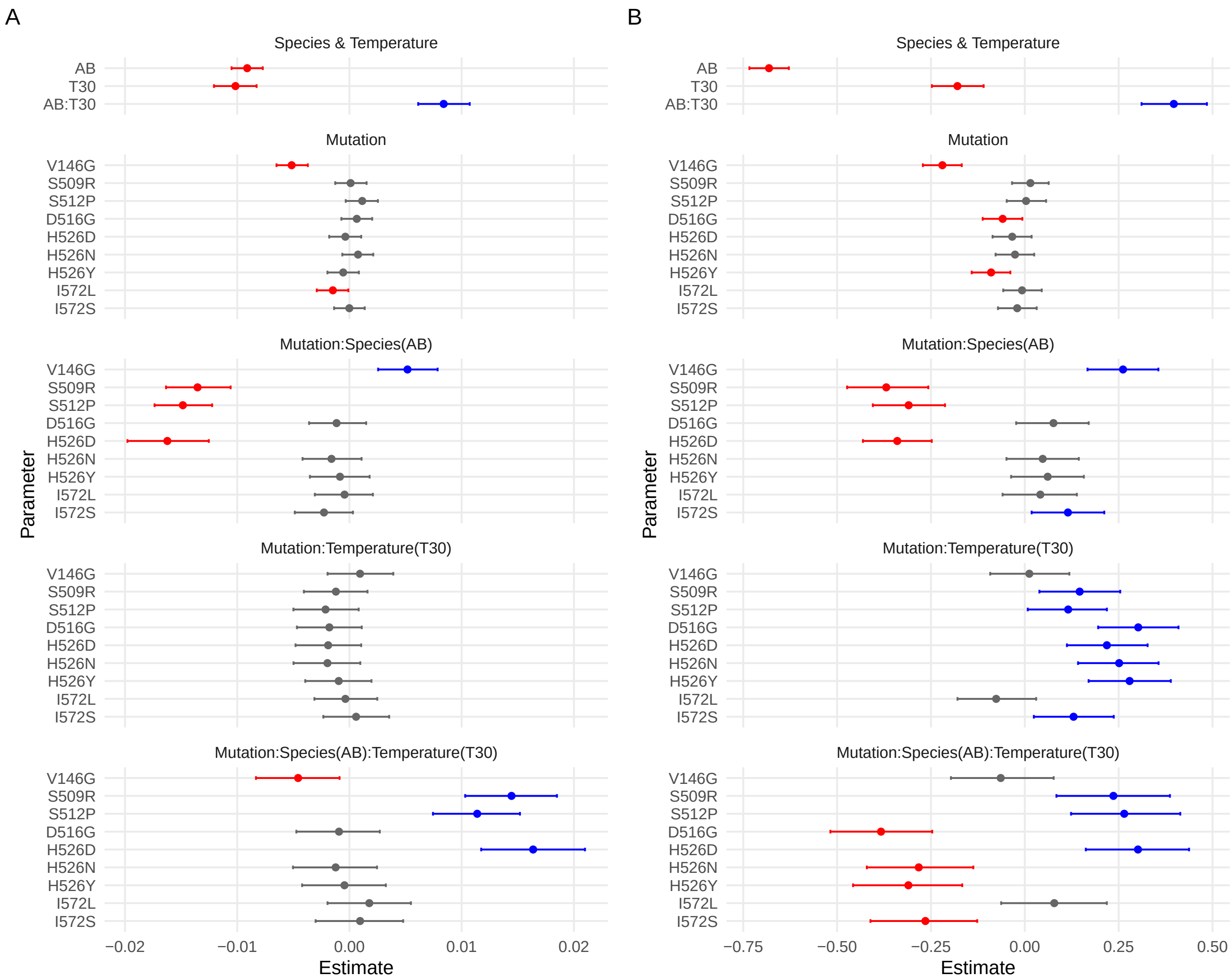

Figure 4: Full model fit to growth data for *E. coli* and *A. baylyi* for A) maximum growth rate and B) maximum OD. Each of the two models includes three explanatory variables (Species, Temperature and Mutation) and all possible interactions between them. The plots show, for each coefficient, the mean of the estimated posterior distribution (dots) and the 95% credible interval (horizontal lines). Coefficients with credible intervals entirely below zero are shown in red, those with credible intervals entirely above zero in blue, and those with credible intervals overlapping with zero are shown in grey. Abbreviations used for coefficients are AB=*A. baylyi* and T30=temperature of 30 °C; colons indicate interactions terms.

One mutation that has been phenotyped for growth in our study (in all three species) and in several previous studies is H526Y. This mutation consistently reduced competitive fitness in *Pseudomonas* but to widely varying degrees, ranging from almost no reduction to almost 27%. Moreover, most of the variation in competitive growth was explained by neither mutation nor species but by epistatic interactions between mutation and species, which in turn were not well explained by phylogenetic distance either (Vogwill et al. 2016). In clinical isolates of *E. coli*, H526Y had a wide range of effects on competitive fitness, with mostly negative effects but in some cases also increasing fitness (Hinz et al. 2024). In *M. tuberculosis*, H526Y causes a fitness reduction of around 20% (Gagneux et al. 2006). Broadly consistent with these results, H526Y reduced maximum growth rate in all three species that we investigated, but only weakly in *E. coli* and *L. plantarum* and without any epistatic interactions between those two species. By contrast, in *A. baylyi*, H526Y induced a reduction of around 16%, with consequently strong epistatic interactions with both other species.

Prior work has also established strong genotype-by-temperature interactions of RIF resistance mutations. Early experiments by Jin and Gross (1989) showed that some mutations in *rpoB* that confer RIF resistance also increased sensitivity to high and/or low temperatures in *E. coli*. Assuming that 37 °C constitutes a mild heat stress in *A. baylyi*, this is in accord with our results that three mutants in this species had very low growth rates and reached very low maximum OD. However, at the level of individual mutation this parallelism does not hold: *E. coli* carrying mutation H526Y were unable to grow at 44 °C in Jin and Gross (1989), but *A. baylyi* carrying the homologous mutation did not exhibit any particular heat sensitivity. This is also in line with experiments in *Pseudomonas aeruginosa* in which competitive growth of a H526Y mutant was not affected by high temperature (Gifford et al. 2016). Interestingly though, one of the three mutations that did induce heat sensitivity in *A. baylyi* in our experiments, H526D, is at the same site as H526Y; unfortunately H526D was not phenotyped in either of these two studies. Another mutation that has previously been reported to induce heat sensitivity, in *Listeria monocytogenes*, was D516G (Morse et al. 1999). Again, the corresponding *A. baylyi* mutant grew very well at 37 °C in our experiments.

In addition to inducing heat-sensitivity, RIF resistance mutations in *rpoB* can also have the opposite effect, i.e. confer resistance to thermal stress. In *E. coli* populations that had been experimentally evolved for 2000 generations at high temperature (42.2 °C) but in the absence of RIF, Rodríguez-Verdugo et al. (2013) observed the the repeated spread of four mutations that substantially increased competitive fitness at 42.2 °C relative to the ancestor. Surprisingly though, these effects were very sensitive to genetic background: in a different *E. coli* strain (MG1655, the strain which we also studied), the same mutations reduced fitness at 42.2 °C. One of these mutations (I572L) overlapped with our panel of mutants that we phenotyped in *A. baylyi* at 37 °C, but this mutation affected maximum growth rate and maximum OD very little relative to the wildtype.

Similar, mutation-specific effects of temperature on fitness also seem to be involved in resistance to other antibiotics (Mira et al. 2022). Moreover, similar to temperature, strong GxE interactions have also be reported between RIF resistance mutations in *E. coli* and different nutrient conditions, including different growth media (Clarke et al. 2020; Soley et al. 2023; Hinz et al. 2024), different carbon sources (Gifford et al. 2016), and elemental nutrient limitation (Maharjan and Ferenci 2017). In many instances in these studies, not only the magnitude but also the direction of mutational fitness effects depended on the available nutrients, mirroring our and previous results on genotype by temperature interactions. Together, these results indicate that GxE interactions are pervasive in antibiotic resistance mutations.

How can we explain the wide range of GxG, GxE and GxGxE effects caused by single point mutations in the *rpoB* gene? Clearly, the RNAP and its beta subunit that *rpoB* codes for are central to the physiology of all bacteria, and as such pleiotropic effects of mutations in this gene that are unrelated to the binding affinity of RIF are to be expected. Mutations in *rpoB* have been shown to both increase or decrease the speed and efficiency of transcription (Qi et al. 2014; Yang et al. 2023). Mutations that increase the transcription speed (fast elongation rate and low pausing frequency) were reported to also rapidly deplete nucleotides within the cell, which in turn made them more sensitive to compounds such as 5FU and the antibiotic trimethoprim. Those mutants also had a growth advantage at low temperature (25 °C) but grew slowly than the wildtype under heat stress (42 °C) (Yang et al. 2023). In turn, differences in elongation rate and other properties of the RNAP appear to have global effects on gene expression within the cell (Qi et al. 2014; Rodríguez-Verdugo et al. 2016; Trauner et al. 2021; Soley et al. 2023). This transcriptional dysregulation has been shown to be highly idiosyncratic, with individual RIF resistance mutations in *rpoB* up- or down-regulating many genes in different ways (Qi et al. 2014; Trauner et al. 2021; Soley et al. 2023). Given that temperature also has large effects on gene expression, mutation by temperature interactions are expected to ensue. Interestingly, some RIF resistance mutations have been reported to largely reverse transcriptional changes brought about by heat stress (Rodríguez-Verdugo et al. 2016). Overall, although we are only beginning to understand the mechanistic basis of fitness effects of RIF resistance mutations, the findings of mutation-specific transcriptional dysregulation that is impacted by temperature provides at least some clues.

In conclusion, our study adds to the growing evidence indicating that cross-species predictions of fitness effects of antibiotic resistance mutations is extremely challenging, due to strong epistatic interactions between individual resistance mutations and genomic background. These problems are further compounded by strong environmental effects, in our case temperature, that give rise to both GxE and GxGxE interactions. As a result, accurate predictions of fitness effects of RIF resistance mutations, based on growth data in other species or other environments alone, appear impossible. It remains to be seen whether mechanistic models that integrate more data and processes—e.g. the effects of *rpoB* mutations on RNAP structure and enzymatic efficiency—can pave a way forward.

Code availability

Code and data for this project will be made available on Github following publication.

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Competing Interests

The authors declare no competing financial interests.

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Supplementary Material

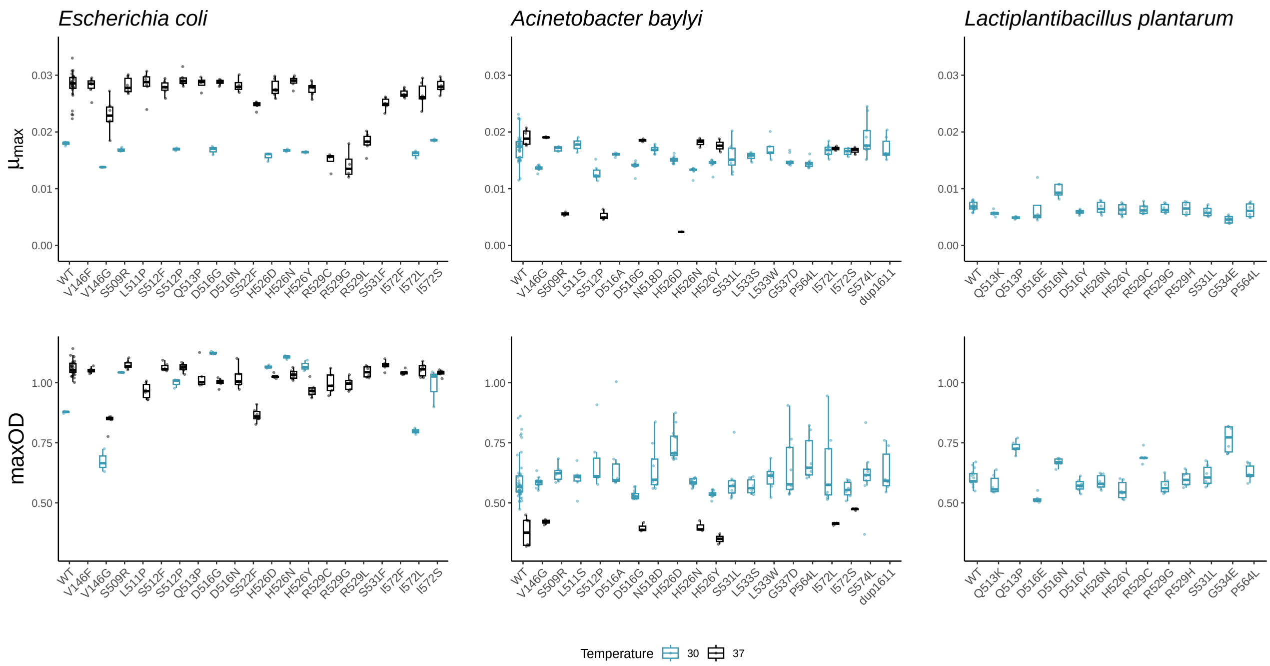

Figure S1: Measured growth traits for all three species, the two temperature conditions and all rifampicin resistant mutants. Small dots indicate estimates from individual growth assays and boxes show the median and interquartile range, with whiskers indicating the highest and lowest value within 1.5 × the interquartile range.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	0.0281	0.0003	0.0275	0.0288
b_MutationD516G	0.0006	0.0008	-0.0009	0.0022
b_MutationH526D	-0.0004	0.0008	-0.0020	0.0013
b_MutationH526N	0.0008	0.0008	-0.0008	0.0024
b_MutationH526Y	-0.0006	0.0008	-0.0021	0.0010
b_MutationI572L	-0.0015	0.0008	-0.0031	0.0001
b_MutationI572S	-0.0000	0.0008	-0.0016	0.0016
b_MutationS509R	0.0001	0.0008	-0.0015	0.0017
b_MutationS512P	0.0011	0.0008	-0.0005	0.0027
b_MutationV146G	-0.0052	0.0008	-0.0067	-0.0035
b_Temperature30	-0.0102	0.0011	-0.0123	-0.0080
b_MutationD516G:Temperature30	-0.0018	0.0017	-0.0050	0.0015
b_MutationH526D:Temperature30	-0.0019	0.0017	-0.0054	0.0014
b_MutationH526N:Temperature30	-0.0020	0.0017	-0.0052	0.0013
b_MutationH526Y:Temperature30	-0.0010	0.0017	-0.0042	0.0024
b_MutationI572L:Temperature30	-0.0004	0.0017	-0.0037	0.0028
b_MutationI572S:Temperature30	0.0006	0.0017	-0.0028	0.0039
b_MutationS509R:Temperature30	-0.0012	0.0017	-0.0047	0.0021
b_MutationS512P:Temperature30	-0.0021	0.0017	-0.0054	0.0011
b_MutationV146G:Temperature30	0.0010	0.0017	-0.0024	0.0044
sigma	0.0018	0.0001	0.0016	0.0021
Intercept	0.0248	0.0002	0.0244	0.0251
lprior	-3.1413	0.0000	-3.1413	-3.1413
lp__	540.1453	3.5393	532.2623	546.1706

Table S1: Summary of the posterior distribution for the GxE model for mumax as the response variable, and mutation and temperature as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species *E. coli*. Formula used for fit: $\mu_{\max} \sim \text{Mutation} * \text{Temperature}$.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	0.0190	0.0006	0.0178	0.0202
b_MutationD516G	-0.0005	0.0011	-0.0026	0.0015
b_MutationH526D	-0.0167	0.0016	-0.0198	-0.0135
b_MutationH526N	-0.0009	0.0010	-0.0029	0.0012
b_MutationH526Y	-0.0014	0.0011	-0.0035	0.0007
b_MutationI572L	-0.0019	0.0010	-0.0040	0.0001
b_MutationI572S	-0.0023	0.0010	-0.0043	-0.0002
b_MutationS509R	-0.0135	0.0012	-0.0159	-0.0111
b_MutationS512P	-0.0137	0.0010	-0.0157	-0.0117
b_MutationV146G	0.0001	0.0011	-0.0020	0.0021
b_Temperature30	-0.0018	0.0006	-0.0030	-0.0006
b_MutationD516G:Temperature30	-0.0027	0.0012	-0.0050	-0.0005
b_MutationH526D:Temperature30	0.0145	0.0017	0.0113	0.0177
b_MutationH526N:Temperature30	-0.0032	0.0011	-0.0054	-0.0010
b_MutationH526Y:Temperature30	-0.0014	0.0012	-0.0037	0.0009
b_MutationI572L:Temperature30	0.0014	0.0012	-0.0008	0.0037
b_MutationI572S:Temperature30	0.0015	0.0013	-0.0011	0.0041
b_MutationS509R:Temperature30	0.0133	0.0014	0.0106	0.0159
b_MutationS512P:Temperature30	0.0093	0.0012	0.0069	0.0117
b_MutationV146G:Temperature30	-0.0037	0.0012	-0.0060	-0.0014
sigma	0.0015	0.0001	0.0013	0.0017
Intercept	0.0155	0.0001	0.0152	0.0157
lprior	-3.1412	0.0000	-3.1412	-3.1412
lp__	826.6984	3.4437	818.9579	832.2821

Table S2: Summary of the posterior distribution for the GxE model for mumax as the response variable, and mutation and temperature as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species *A. baylyi*. Formula used for fit: $\mu_{\max} \sim$ Mutation $*$ Temperature.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	1.0599	0.0052	1.0498	1.0704
b_MutationD516G	-0.0587	0.0123	-0.0821	-0.0345
b_MutationH526D	-0.0333	0.0125	-0.0573	-0.0091
b_MutationH526N	-0.0257	0.0125	-0.0497	-0.0009
b_MutationH526Y	-0.0893	0.0123	-0.1132	-0.0648
b_MutationI572L	-0.0071	0.0123	-0.0320	0.0176
b_MutationI572S	-0.0207	0.0122	-0.0451	0.0033
b_MutationS509R	0.0141	0.0123	-0.0105	0.0379
b_MutationS512P	0.0027	0.0125	-0.0214	0.0274
b_MutationV146G	-0.2204	0.0127	-0.2467	-0.1966
b_Temperature30	-0.1832	0.0164	-0.2147	-0.1515
b_MutationD516G:Temperature30	0.3061	0.0248	0.2573	0.3536
b_MutationH526D:Temperature30	0.2226	0.0252	0.1709	0.2717
b_MutationH526N:Temperature30	0.2549	0.0252	0.2051	0.3031
b_MutationH526Y:Temperature30	0.2820	0.0256	0.2310	0.3314
b_MutationI572L:Temperature30	-0.0713	0.0250	-0.1207	-0.0230
b_MutationI572S:Temperature30	0.1340	0.0252	0.0841	0.1849
b_MutationS509R:Temperature30	0.1515	0.0248	0.1029	0.1998
b_MutationS512P:Temperature30	0.1198	0.0256	0.0701	0.1704
b_MutationV146G:Temperature30	0.0171	0.0255	-0.0327	0.0678
sigma	0.0274	0.0021	0.0236	0.0318
Intercept	1.0140	0.0027	1.0087	1.0191
lprior	-3.1413	0.0000	-3.1413	-3.1413
lp__	240.3367	3.4888	232.5543	246.1141

Table S3: Summary of the posterior distribution for the GxE model for maxOD as the response variable, and mutation and temperature as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species *E. coli.* Formula used for fit: maxOD $\sim$ Mutation $*$ Temperature.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	0.3784	0.0295	0.3222	0.4355
b_MutationD516G	0.0181	0.0502	-0.0835	0.1169
b_MutationH526D	-0.3734	0.0462	-0.4654	-0.2849
b_MutationH526N	0.0222	0.0501	-0.0770	0.1199
b_MutationH526Y	-0.0294	0.0498	-0.1252	0.0678
b_MutationI572L	0.0351	0.0506	-0.0621	0.1338
b_MutationI572S	0.0938	0.0485	-0.0045	0.1876
b_MutationS509R	-0.3538	0.0574	-0.4671	-0.2400
b_MutationS512P	-0.3044	0.0503	-0.4044	-0.2070
b_MutationV146G	0.0428	0.0503	-0.0553	0.1418
b_Temperature30	0.2175	0.0312	0.1567	0.2779
b_MutationD516G:Temperature30	-0.0815	0.0543	-0.1884	0.0294
b_MutationH526D:Temperature30	0.5200	0.0510	0.4223	0.6193
b_MutationH526N:Temperature30	-0.0311	0.0550	-0.1366	0.0765
b_MutationH526Y:Temperature30	-0.0300	0.0540	-0.1353	0.0749
b_MutationI572L:Temperature30	0.0016	0.0564	-0.1075	0.1078
b_MutationI572S:Temperature30	-0.1340	0.0570	-0.2474	-0.0210
b_MutationS509R:Temperature30	0.3817	0.0656	0.2562	0.5084
b_MutationS512P:Temperature30	0.3797	0.0584	0.2664	0.4932
b_MutationV146G:Temperature30	-0.0527	0.0548	-0.1594	0.0544
sigma	0.0700	0.0042	0.0625	0.0789
Intercept	0.5436	0.0053	0.5333	0.5540
lprior	-3.1421	0.0001	-3.1423	-3.1419
lp__	214.2521	3.5621	206.2013	220.0754

Table S4: Summary of the posterior distribution for the GxE model for maxOD as the response variable, and mutation and temperature as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species *A. baylyi*. Formula used for fit: maxOD $\sim$ Mutation $*$ Temperature.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	0.0281	0.0003	0.0274	0.0288
b_MutationD516G	0.0006	0.0008	-0.0009	0.0022
b_MutationH526D	-0.0004	0.0008	-0.0019	0.0012
b_MutationH526N	0.0008	0.0008	-0.0007	0.0023
b_MutationH526Y	-0.0005	0.0008	-0.0020	0.0010
b_MutationI572L	-0.0015	0.0008	-0.0031	0.0002
b_MutationI572S	-0.0000	0.0008	-0.0015	0.0016
b_MutationS509R	0.0001	0.0008	-0.0015	0.0016
b_MutationS512P	0.0012	0.0008	-0.0004	0.0027
b_MutationV146G	-0.0051	0.0008	-0.0067	-0.0036
b_SpeciesAbaylyi	-0.0109	0.0004	-0.0117	-0.0100
b_MutationD516G:SpeciesAbaylyi	-0.0039	0.0009	-0.0057	-0.0020
b_MutationH526D:SpeciesAbaylyi	-0.0018	0.0010	-0.0037	0.0002
b_MutationH526N:SpeciesAbaylyi	-0.0048	0.0010	-0.0067	-0.0029
b_MutationH526Y:SpeciesAbaylyi	-0.0023	0.0009	-0.0041	-0.0005
b_MutationI572L:SpeciesAbaylyi	0.0010	0.0010	-0.0011	0.0030
b_MutationI572S:SpeciesAbaylyi	-0.0008	0.0012	-0.0032	0.0016
b_MutationS509R:SpeciesAbaylyi	-0.0003	0.0011	-0.0025	0.0020
b_MutationS512P:SpeciesAbaylyi	-0.0056	0.0011	-0.0078	-0.0034
b_MutationV146G:SpeciesAbaylyi	0.0015	0.0010	-0.0003	0.0034
sigma	0.0017	0.0001	0.0016	0.0019
Intercept	0.0201	0.0001	0.0199	0.0204
lprior	-3.1413	0.0000	-3.1413	-3.1413
lp__	1058.2413	3.4551	1050.6565	1063.8246

Table S5: Summary of the posterior distribution for the GxG model for mumax as the response variable, and mutation and species as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species pair *E. coli* / *A. baylyi*. Formula used for fit: $\mu_{\max} \sim \text{Mutation} * \text{Species}$.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	0.0281	0.0003	0.0275	0.0287
b_MutationD516N	0.0001	0.0009	-0.0017	0.0018
b_MutationH526N	0.0007	0.0007	-0.0008	0.0022
b_MutationH526Y	-0.0005	0.0008	-0.0020	0.0009
b_MutationQ513P	0.0004	0.0009	-0.0013	0.0022
b_MutationR529C	-0.0132	0.0009	-0.0149	-0.0115
b_MutationR529G	-0.0139	0.0009	-0.0156	-0.0121
b_SpeciesLplantarum	-0.0212	0.0005	-0.0222	-0.0202
b_MutationD516N:SpeciesLplantarum	0.0025	0.0013	0.0001	0.0051
b_MutationH526N:SpeciesLplantarum	-0.0010	0.0011	-0.0031	0.0011
b_MutationH526Y:SpeciesLplantarum	-0.0001	0.0011	-0.0023	0.0021
b_MutationQ513P:SpeciesLplantarum	-0.0025	0.0012	-0.0048	-0.0002
b_MutationR529C:SpeciesLplantarum	0.0126	0.0012	0.0103	0.0150
b_MutationR529G:SpeciesLplantarum	0.0135	0.0012	0.0111	0.0157
sigma	0.0017	0.0001	0.0014	0.0019
Intercept	0.0170	0.0002	0.0167	0.0173
lprior	-3.1412	0.0000	-3.1412	-3.1412
lp__	518.3420	2.8872	511.7350	522.9177

Table S6: Summary of the posterior distribution for the GxG model for mumax as the response variable, and mutation and species as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species pair *E. coli* / *L. plantarum*. Formula used for fit: $\mu_{\max} \sim \text{Mutation} * \text{Species}$.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	0.0172	0.0003	0.0167	0.0177
b_MutationH526N	-0.0040	0.0005	-0.0051	-0.0030
b_MutationH526Y	-0.0028	0.0005	-0.0039	-0.0018
b_MutationP564L	-0.0028	0.0006	-0.0040	-0.0015
b_MutationS531L	-0.0016	0.0007	-0.0029	-0.0003
b_SpeciesLplantarum	-0.0103	0.0005	-0.0113	-0.0093
b_MutationH526N:SpeciesLplantarum	0.0037	0.0010	0.0018	0.0055
b_MutationH526Y:SpeciesLplantarum	0.0022	0.0010	0.0002	0.0042
b_MutationP564L:SpeciesLplantarum	0.0021	0.0010	0.0000	0.0041
b_MutationS531L:SpeciesLplantarum	0.0006	0.0011	-0.0015	0.0027
sigma	0.0017	0.0001	0.0015	0.0020
Intercept	0.0130	0.0002	0.0127	0.0133
lprior	-3.1412	0.0000	-3.1412	-3.1412
lp__	624.0497	2.5026	617.9604	627.8308

Table S7: Summary of the posterior distribution for the GxG model for mumax as the response variable, and mutation and species as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species pair *A. baylyi / L. plantarum*. Formula used for fit: $\mu_{\max} \sim \text{Mutation} * \text{Species}$.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	1.0597	0.0113	1.0371	1.0818
b_MutationD516G	-0.0585	0.0276	-0.1137	-0.0057
b_MutationH526D	-0.0335	0.0275	-0.0862	0.0213
b_MutationH526N	-0.0250	0.0281	-0.0803	0.0305
b_MutationH526Y	-0.0891	0.0279	-0.1434	-0.0335
b_MutationI572L	-0.0071	0.0276	-0.0618	0.0465
b_MutationI572S	-0.0203	0.0271	-0.0748	0.0333
b_MutationS509R	0.0145	0.0280	-0.0400	0.0684
b_MutationS512P	0.0030	0.0283	-0.0520	0.0570
b_MutationV146G	-0.2207	0.0283	-0.2758	-0.1658
b_SpeciesAbaylyi	-0.4638	0.0144	-0.4920	-0.4360
b_MutationD516G:SpeciesAbaylyi	-0.0046	0.0327	-0.0671	0.0602
b_MutationH526D:SpeciesAbaylyi	0.1801	0.0335	0.1147	0.2452
b_MutationH526N:SpeciesAbaylyi	0.0159	0.0339	-0.0507	0.0827
b_MutationH526Y:SpeciesAbaylyi	0.0296	0.0338	-0.0364	0.0970
b_MutationI572L:SpeciesAbaylyi	0.0432	0.0356	-0.0243	0.1138
b_MutationI572S:SpeciesAbaylyi	-0.0202	0.0376	-0.0925	0.0550
b_MutationS509R:SpeciesAbaylyi	0.0140	0.0384	-0.0584	0.0903
b_MutationS512P:SpeciesAbaylyi	0.0713	0.0395	-0.0066	0.1465
b_MutationV146G:SpeciesAbaylyi	0.2105	0.0338	0.1449	0.2765
sigma	0.0619	0.0031	0.0564	0.0681
Intercept	0.7561	0.0041	0.7480	0.7643
lprior	-3.1442	0.0001	-3.1445	-3.1440
lp__	304.1830	3.4085	296.4586	309.8772

Table S8: Summary of the posterior distribution for the GxG model for maxOD as the response variable, and mutation and species as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species pair *E. coli* / *A. baylyi*. Formula used for fit: maxOD $\sim$ Mutation $*$ Species.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	1.0598	0.0064	1.0474	1.0724
b_MutationD516N	-0.0383	0.0166	-0.0702	-0.0058
b_MutationH526N	-0.0255	0.0154	-0.0554	0.0044
b_MutationH526Y	-0.0892	0.0155	-0.1188	-0.0570
b_MutationQ513P	-0.0325	0.0165	-0.0650	0.0001
b_MutationR529C	-0.0607	0.0166	-0.0925	-0.0277
b_MutationR529G	-0.0651	0.0169	-0.0977	-0.0318
b_SpeciesLplantarum	-0.4554	0.0109	-0.4768	-0.4340
b_MutationD516N:SpeciesLplantarum	0.1025	0.0234	0.0559	0.1490
b_MutationH526N:SpeciesLplantarum	0.0078	0.0225	-0.0366	0.0512
b_MutationH526Y:SpeciesLplantarum	0.0368	0.0226	-0.0093	0.0800
b_MutationQ513P:SpeciesLplantarum	0.1590	0.0230	0.1142	0.2049
b_MutationR529C:SpeciesLplantarum	0.1478	0.0237	0.1008	0.1937
b_MutationR529G:SpeciesLplantarum	0.0317	0.0239	-0.0150	0.0783
sigma	0.0340	0.0025	0.0296	0.0392
Intercept	0.8469	0.0031	0.8406	0.8530
lprior	-3.1438	0.0001	-3.1440	-3.1436
lp__	213.9886	2.8610	207.7304	218.7550

Table S9: Summary of the posterior distribution for the GxG model for maxOD as the response variable, and mutation and species as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species pair *E. coli* / *L. plantarum.* Formula used for fit: maxOD $\sim$ Mutation $*$ Species.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	0.5958	0.0096	0.5770	0.6144
b_MutationH526N	-0.0087	0.0206	-0.0488	0.0327
b_MutationH526Y	-0.0596	0.0203	-0.0990	-0.0205
b_MutationP564L	0.0886	0.0245	0.0419	0.1371
b_MutationS531L	-0.0104	0.0248	-0.0595	0.0379
b_SpeciesLplantarum	0.0087	0.0200	-0.0317	0.0474
b_MutationH526N:SpeciesLplantarum	-0.0094	0.0384	-0.0870	0.0678
b_MutationH526Y:SpeciesLplantarum	0.0072	0.0394	-0.0688	0.0852
b_MutationP564L:SpeciesLplantarum	-0.0671	0.0405	-0.1451	0.0123
b_MutationS531L:SpeciesLplantarum	0.0202	0.0422	-0.0651	0.1004
sigma	0.0687	0.0044	0.0608	0.0782
Intercept	0.5947	0.0059	0.5831	0.6062
lprior	-3.1417	0.0001	-3.1419	-3.1416
lp__	163.4796	2.5057	157.4632	167.2444

Table S10: Summary of the posterior distribution for the GxG model for maxOD as the response variable, and mutation and species as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species pair *A. baylyi* / *L. plantarum*. Formula used for fit: maxOD $\sim$ Mutation $*$ Species.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	0.0281	0.0003	0.0275	0.0287
b_MutationD516G	0.0007	0.0007	-0.0007	0.0020
b_MutationH526D	-0.0004	0.0007	-0.0018	0.0010
b_MutationH526N	0.0008	0.0007	-0.0006	0.0021
b_MutationH526Y	-0.0006	0.0007	-0.0020	0.0008
b_MutationI572L	-0.0015	0.0007	-0.0029	-0.0001
b_MutationI572S	-0.0000	0.0007	-0.0014	0.0014
b_MutationS509R	0.0001	0.0007	-0.0013	0.0015
b_MutationS512P	0.0011	0.0007	-0.0003	0.0025
b_MutationV146G	-0.0051	0.0007	-0.0065	-0.0037
b_SpeciesAbaylyi	-0.0091	0.0007	-0.0105	-0.0077
b_Temperature30	-0.0102	0.0010	-0.0121	-0.0083
b_MutationD516G:SpeciesAbaylyi	-0.0011	0.0013	-0.0036	0.0015
b_MutationH526D:SpeciesAbaylyi	-0.0162	0.0019	-0.0198	-0.0125
b_MutationH526N:SpeciesAbaylyi	-0.0016	0.0013	-0.0042	0.0011
b_MutationH526Y:SpeciesAbaylyi	-0.0008	0.0014	-0.0035	0.0018
b_MutationI572L:SpeciesAbaylyi	-0.0004	0.0013	-0.0031	0.0021
b_MutationI572S:SpeciesAbaylyi	-0.0023	0.0013	-0.0049	0.0003
b_MutationS509R:SpeciesAbaylyi	-0.0135	0.0015	-0.0163	-0.0106
b_MutationS512P:SpeciesAbaylyi	-0.0148	0.0013	-0.0174	-0.0122
b_MutationV146G:SpeciesAbaylyi	0.0052	0.0014	0.0026	0.0079
b_MutationD516G:Temperature30	-0.0018	0.0015	-0.0047	0.0011
b_MutationH526D:Temperature30	-0.0019	0.0015	-0.0048	0.0010
b_MutationH526N:Temperature30	-0.0020	0.0015	-0.0050	0.0010
b_MutationH526Y:Temperature30	-0.0010	0.0015	-0.0039	0.0020
b_MutationI572L:Temperature30	-0.0004	0.0014	-0.0031	0.0025
b_MutationI572S:Temperature30	0.0006	0.0015	-0.0023	0.0035
b_MutationS509R:Temperature30	-0.0012	0.0015	-0.0041	0.0016
b_MutationS512P:Temperature30	-0.0021	0.0015	-0.0050	0.0008
b_MutationV146G:Temperature30	0.0009	0.0015	-0.0020	0.0039
b_SpeciesAbaylyi:Temperature30	0.0084	0.0012	0.0061	0.0107
b_MutationD516G:SpeciesAbaylyi:Temperature30	-0.0009	0.0019	-0.0047	0.0027
b_MutationH526D:SpeciesAbaylyi:Temperature30	0.0164	0.0023	0.0117	0.0210
b_MutationH526N:SpeciesAbaylyi:Temperature30	-0.0012	0.0019	-0.0050	0.0025
b_MutationH526Y:SpeciesAbaylyi:Temperature30	-0.0004	0.0019	-0.0042	0.0032
b_MutationI572L:SpeciesAbaylyi:Temperature30	0.0018	0.0019	-0.0020	0.0055
b_MutationI572S:SpeciesAbaylyi:Temperature30	0.0009	0.0020	-0.0030	0.0048
b_MutationS509R:SpeciesAbaylyi:Temperature30	0.0144	0.0021	0.0103	0.0185
b_MutationS512P:SpeciesAbaylyi:Temperature30	0.0114	0.0020	0.0074	0.0152
b_MutationV146G:SpeciesAbaylyi:Temperature30	-0.0046	0.0019	-0.0083	-0.0009
sigma	0.0016	0.0001	0.0015	0.0018
Intercept	0.0192	0.0001	0.0190	0.0194
lprior	-3.1413	0.0000	-3.1413	-3.1413
lp__	1375.6707	4.7964	1365.0407	1383.9234

Table S11: Summary of the posterior distribution for the GxGxE model for mumax as the response variable, and mutation, temperature and species as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species pair *E. coli* / *A. baylyi*. Formula used for fit: $\mu_{\max} \sim \text{Mutation} * \text{Species} * \text{Temperature}$.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	1.0598	0.0108	1.0391	1.0804
b_MutationD516G	-0.0589	0.0266	-0.1121	-0.0066
b_MutationH526D	-0.0335	0.0261	-0.0859	0.0182
b_MutationH526N	-0.0260	0.0258	-0.0778	0.0250
b_MutationH526Y	-0.0897	0.0261	-0.1414	-0.0385
b_MutationI572L	-0.0073	0.0260	-0.0574	0.0451
b_MutationI572S	-0.0203	0.0264	-0.0715	0.0318
b_MutationS509R	0.0149	0.0253	-0.0340	0.0636
b_MutationS512P	0.0034	0.0267	-0.0481	0.0567
b_MutationV146G	-0.2195	0.0261	-0.2715	-0.1680
b_SpeciesAbaylyi	-0.6813	0.0267	-0.7336	-0.6285
b_Temperature30	-0.1794	0.0350	-0.2475	-0.1096
b_MutationD516G:SpeciesAbaylyi	0.0764	0.0487	-0.0229	0.1703
b_MutationH526D:SpeciesAbaylyi	-0.3397	0.0465	-0.4312	-0.2479
b_MutationH526N:SpeciesAbaylyi	0.0477	0.0489	-0.0488	0.1439
b_MutationH526Y:SpeciesAbaylyi	0.0610	0.0485	-0.0365	0.1573
b_MutationI572L:SpeciesAbaylyi	0.0414	0.0496	-0.0591	0.1387
b_MutationI572S:SpeciesAbaylyi	0.1149	0.0496	0.0183	0.2118
b_MutationS509R:SpeciesAbaylyi	-0.3690	0.0540	-0.4733	-0.2570
b_MutationS512P:SpeciesAbaylyi	-0.3094	0.0496	-0.4048	-0.2126
b_MutationV146G:SpeciesAbaylyi	0.2616	0.0486	0.1670	0.3559
b_MutationD516G:Temperature30	0.3022	0.0540	0.1954	0.4095
b_MutationH526D:Temperature30	0.2186	0.0549	0.1122	0.3272
b_MutationH526N:Temperature30	0.2513	0.0539	0.1419	0.3563
b_MutationH526Y:Temperature30	0.2791	0.0555	0.1700	0.3890
b_MutationI572L:Temperature30	-0.0763	0.0532	-0.1793	0.0303
b_MutationI572S:Temperature30	0.1299	0.0543	0.0241	0.2369
b_MutationS509R:Temperature30	0.1461	0.0543	0.0390	0.2542
b_MutationS512P:Temperature30	0.1156	0.0538	0.0079	0.2187
b_MutationV146G:Temperature30	0.0121	0.0539	-0.0921	0.1187
b_SpeciesAbaylyi:Temperature30	0.3969	0.0437	0.3110	0.4851
b_MutationD516G:SpeciesAbaylyi:Temperature30	-0.3830	0.0703	-0.5178	-0.2466
b_MutationH526D:SpeciesAbaylyi:Temperature30	0.3015	0.0704	0.1625	0.4375
b_MutationH526N:SpeciesAbaylyi:Temperature30	-0.2825	0.0702	-0.4208	-0.1372
b_MutationH526Y:SpeciesAbaylyi:Temperature30	-0.3101	0.0716	-0.4575	-0.1666
b_MutationI572L:SpeciesAbaylyi:Temperature30	0.0785	0.0716	-0.0630	0.2185
b_MutationI572S:SpeciesAbaylyi:Temperature30	-0.2648	0.0734	-0.4110	-0.1268
b_MutationS509R:SpeciesAbaylyi:Temperature30	0.2360	0.0760	0.0842	0.3866
b_MutationS512P:SpeciesAbaylyi:Temperature30	0.2649	0.0732	0.1232	0.4141
b_MutationV146G:SpeciesAbaylyi:Temperature30	-0.0640	0.0699	-0.1967	0.0769
sigma	0.0578	0.0025	0.0531	0.0631
Intercept	0.7268	0.0033	0.7204	0.7335
lprior	-3.1433	0.0001	-3.1435	-3.1431
lp__	409.8764	4.8881	399.2445	418.3950

Table S12: Summary of the posterior distribution for the GxGxE model for maxOD as the response variable, and mutation, temperature and species as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species pair *E. coli* / *A. baylyi*. Formula used for fit: maxOD $\sim$ Mutation $*$ Species $*$ Temperature.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	0.0283	0.0004	0.0276	0.0291
b_MutationD516G	0.0011	0.0009	-0.0006	0.0028
b_MutationH526D	-0.0020	0.0008	-0.0037	-0.0004
b_MutationH526N	0.0013	0.0008	-0.0003	0.0029
b_MutationH526Y	-0.0002	0.0008	-0.0018	0.0015
b_MutationI572L	-0.0015	0.0009	-0.0032	0.0002
b_MutationI572S	0.0001	0.0008	-0.0016	0.0018
b_MutationS509R	-0.0017	0.0009	-0.0034	-0.0000
b_MutationS512P	-0.0005	0.0009	-0.0022	0.0012
b_MutationV146G	-0.0040	0.0008	-0.0057	-0.0023
b_SpeciesAbaylyi	-0.0101	0.0008	-0.0116	-0.0085
b_Temperature30	-0.0120	0.0008	-0.0136	-0.0103
b_MutationD516G:SpeciesAbaylyi	-0.0021	0.0012	-0.0045	0.0004
b_MutationH526D:SpeciesAbaylyi	-0.0052	0.0014	-0.0079	-0.0025
b_MutationH526N:SpeciesAbaylyi	-0.0028	0.0012	-0.0052	-0.0004
b_MutationH526Y:SpeciesAbaylyi	-0.0016	0.0012	-0.0040	0.0008
b_MutationI572L:SpeciesAbaylyi	0.0001	0.0013	-0.0025	0.0026
b_MutationI572S:SpeciesAbaylyi	-0.0021	0.0013	-0.0047	0.0004
b_MutationS509R:SpeciesAbaylyi	-0.0060	0.0013	-0.0086	-0.0035
b_MutationS512P:SpeciesAbaylyi	-0.0095	0.0013	-0.0121	-0.0070
b_MutationV146G:SpeciesAbaylyi	0.0021	0.0012	-0.0003	0.0045
b_SpeciesAbaylyi:Temperature30	0.0111	0.0006	0.0098	0.0124
b_MutationD516G:Temperature30	-0.0020	0.0012	-0.0044	0.0005
b_MutationH526D:Temperature30	0.0043	0.0014	0.0015	0.0071
b_MutationH526N:Temperature30	-0.0023	0.0012	-0.0046	0.0001
b_MutationH526Y:Temperature30	-0.0009	0.0012	-0.0033	0.0015
b_MutationI572L:Temperature30	0.0010	0.0013	-0.0014	0.0035
b_MutationI572S:Temperature30	0.0016	0.0013	-0.0009	0.0041
b_MutationS509R:Temperature30	0.0055	0.0014	0.0029	0.0082
b_MutationS512P:Temperature30	0.0040	0.0013	0.0015	0.0065
b_MutationV146G:Temperature30	-0.0012	0.0012	-0.0036	0.0012
sigma	0.0021	0.0001	0.0019	0.0022
Intercept	0.0192	0.0001	0.0190	0.0195
lprior	-3.1413	0.0000	-3.1413	-3.1413
lp__	1306.9011	4.1754	1297.8630	1314.1786

Table S13: Summary of the posterior distribution for the GxGxE model for mumax as the response variable, and mutation, temperature and species as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species pair *E. coli* / *A. baylyi*. Formula used for fit: $\mu_{\max} \sim \text{Mutation} * \text{Species} + \text{Temperature} * \text{Species} + \text{Mutation} * \text{Temperature}$.

	Estimate	Est.Error	Q2.5	Q97.5
b_Intercept	1.0569	0.0140	1.0290	1.0837
b_MutationD516G	0.0071	0.0313	-0.0540	0.0681
b_MutationH526D	-0.0991	0.0313	-0.1624	-0.0392
b_MutationH526N	0.0219	0.0310	-0.0400	0.0829
b_MutationH526Y	-0.0372	0.0317	-0.0981	0.0245
b_MutationI572L	-0.0251	0.0311	-0.0857	0.0359
b_MutationI572S	0.0191	0.0321	-0.0429	0.0834
b_MutationS509R	-0.0224	0.0325	-0.0858	0.0409
b_MutationS512P	-0.0452	0.0318	-0.1089	0.0174
b_MutationV146G	-0.2124	0.0316	-0.2736	-0.1495
b_SpeciesAbaylyi	-0.6644	0.0276	-0.7183	-0.6113
b_Temperature30	-0.1514	0.0306	-0.2126	-0.0914
b_MutationD516G:SpeciesAbaylyi	-0.1283	0.0444	-0.2183	-0.0413
b_MutationH526D:SpeciesAbaylyi	-0.1849	0.0435	-0.2688	-0.1003
b_MutationH526N:SpeciesAbaylyi	-0.1017	0.0436	-0.1854	-0.0185
b_MutationH526Y:SpeciesAbaylyi	-0.1031	0.0452	-0.1897	-0.0130
b_MutationI572L:SpeciesAbaylyi	0.0886	0.0458	-0.0012	0.1783
b_MutationI572S:SpeciesAbaylyi	-0.0124	0.0464	-0.1006	0.0796
b_MutationS509R:SpeciesAbaylyi	-0.2264	0.0484	-0.3210	-0.1281
b_MutationS512P:SpeciesAbaylyi	-0.1716	0.0464	-0.2623	-0.0825
b_MutationV146G:SpeciesAbaylyi	0.2332	0.0448	0.1468	0.3257
b_SpeciesAbaylyi:Temperature30	0.3530	0.0226	0.3086	0.3978
b_MutationD516G:Temperature30	0.0868	0.0444	0.0007	0.1755
b_MutationH526D:Temperature30	0.3979	0.0437	0.3139	0.4842
b_MutationH526N:Temperature30	0.0910	0.0446	0.0039	0.1760
b_MutationH526Y:Temperature30	0.1032	0.0448	0.0141	0.1901
b_MutationI572L:Temperature30	-0.0397	0.0452	-0.1287	0.0514
b_MutationI572S:Temperature30	-0.0086	0.0459	-0.1023	0.0793
b_MutationS509R:Temperature30	0.2384	0.0505	0.1389	0.3370
b_MutationS512P:Temperature30	0.2416	0.0466	0.1510	0.3331
b_MutationV146G:Temperature30	-0.0273	0.0451	-0.1174	0.0614
sigma	0.0757	0.0033	0.0694	0.0824
Intercept	0.7269	0.0046	0.7178	0.7363
lprior	-3.1435	0.0001	-3.1438	-3.1433
lp__	332.2377	4.4018	322.8632	339.6354

Table S14: Summary of the posterior distribution for the GxGxE model for maxOD as the response variable, and mutation, temperature and species as independent variables. The columns show the mean, standard deviation, and credible interval of the posterior distribution for each estimated parameter. This model fit is for species pair *E. coli* / *A. baylyi*. Formula used for fit: maxOD $\sim$ Mutation $*$ Species + Temperature $*$ Species + Mutation $*$ Temperature.